INCH-POUND

MIL-S-20708/67C
13 March 1989

SUPERSEDING
MIL-S-20708/67B
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, TORQUE DIFFERENTIAL RECEIVER, TYPE 31TDR6B

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I. Requirements.

Requirement	Value		Unit	Tolerance	
Frequency	60	©	Hz	±1%	
Primary Voltage	78	©	volts	±1%	
Primary Current	748.0		milliamps	maximum	
Primary Power	7.8		watts	maximum	
Impedance					
Zp	----		ohms	min.-max.	
Zd	----		ohms	min.-max.	
Impedance Angle:					
Zp	----		degrees	min.-max.	
Zd	----		degrees	min.-max.	
Transformation Ratio	1.154		------	±2%	
Phase Shift (Lead)	6.5		degrees	maximum	
Electrical Error	10.0		minutes	maximum	
Receiver Error	48.0		minutes	maximum	
Synchronizing Time: 30° ±2°	1.0		seconds	maximum	
Synchronizing Time: 177° ±2°	2.0		seconds	maximum	
Torque Gradient	0.30		ounce-inches per degree	minimum	
Radial Play	0.0008		inches	maximum	
End Play	0.0030	©	inches	±0.0020	©
Temperature Rise	18.0		degrees (C)	maximum	
Variation of Voltage (+10%) and Freq. (-5%)	----		watts	maximum	

© denotes changes

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TABLE II. Military part number variant characteristics.

Military Part No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U +.0000 −.0002
M20708/67-01B	.700	.730	3.635	Splined	Terminal Board	.2405

1/Military part number M20708/67-01B shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Military part number cross-references.

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/67A	MIL-S-20708/67	
M20708/67A-001	M20708/67-001	M20708/67-01B

NOTE: All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein.

Custodians:
Navy-AS

Review Activities:
DLA-ES

User Activities:

Preparing Activity:
Navy-AS

(Project 5990-0349-41)

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